

Notes: Diamond (RES, 1984)

Financial Intermediation and Delegated Monitoring

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1 Big picture

- **Question.** Why do financial intermediaries exist if depositors could in principle lend directly to firms? More precisely: under what conditions is it efficient to **delegate monitoring** of borrowers to an intermediary that is *itself* not directly monitored by depositors?
- **Setting.** The paper studies a world with **entrepreneurs** who need outside finance, **many lenders/depositors**, and a **costly monitoring technology** that reveals project outcomes useful for enforcing repayment. The key candidate intermediary is a bank-like institution that borrows from depositors, lends to firms, and monitors firms on depositors' behalf.
- **Core challenge.** Delegation solves one incentive problem but creates another. If depositors delegate monitoring to an intermediary, the intermediary can underreport what it collected from borrowers or fail to monitor properly. So one must compare the **gross benefits** of intermediation with the **delegation costs** of making the intermediary honest.
- **Main theoretical contribution.** The paper gives one of the classic formal theories of banking as **delegated monitoring**. It explicitly characterizes the incentive costs of delegation and shows how **diversification inside the intermediary** can make those costs arbitrarily small.
- **Main mechanism.** If an intermediary monitors many independent loans, its aggregate cash flow becomes more stable. Then a simple fixed promised payment to depositors can be made incentive compatible with very little expected bankruptcy cost. Diversification is therefore what makes delegated monitoring viable.
- **Why it matters.** The paper explains several stylized facts about banks and intermediaries:
 - they are typically **highly leveraged**,
 - they issue mostly **safe debt-like claims** to depositors,
 - they hold **illiquid private loans**, and
 - they benefit from **scale and diversification**.
- **Main results.**
 - The optimal contract between an entrepreneur and lenders is a **debt contract with bankruptcy penalties**.
 - Direct monitoring by many lenders is wasteful because each lender must either duplicate monitoring effort or free ride.
 - An intermediary that monitors only **one** borrower is not viable: it inherits essentially the same incentive problem as the borrower, and adds monitoring costs on top.
 - When the intermediary monitors **many independent borrowers**, the expected delegation cost per borrower, D_N , declines with portfolio size and converges to **zero** as $N \rightarrow \infty$.
 - As diversification grows, the intermediary's optimal liability structure becomes mostly **debt/deposits**, and that debt becomes **almost riskless**.
 - In the risk-aversion extension, diversification by **subdividing risks** helps because it improves risk sharing, while diversification by merely **adding independent risks** helps only under special utility conditions.

- In a Leland–Pyle signalling environment, intermediation is viable when bankers can **subdivide risks**, but not when a single banker simply accumulates more independent risks under constant absolute risk aversion.
- **Economic takeaway.** Banks are not just large pools of capital. Their essential role is organizational: they centralize costly monitoring and make that delegation incentive compatible by holding a **diversified portfolio** and issuing mostly **fixed debt claims**.

2 Data and measurement (key objects)

2.1 This is a theory paper: what the key objects are

- There is **no empirical dataset**. The paper is a pure theory paper.
- The key objects are:
 - entrepreneurs and projects,
 - lenders/depositors,
 - the intermediary,
 - project returns y_i ,
 - monitoring cost K ,
 - promised repayment R ,
 - bankruptcy penalties,
 - intermediary delegation cost D_N , and
 - portfolio size N .
- The entire paper asks how these primitives determine whether **direct finance**, **direct monitoring**, or **delegated monitoring** is the cheapest feasible arrangement.

2.2 Agents and market structure

- **Entrepreneurs.**
 - There are N entrepreneurs, indexed by $i = 1, \dots, N$.
 - Each entrepreneur has access to an indivisible project but no internal funds.
- **Lenders / depositors.**
 - Each lender has wealth $1/m$, so a project of size 1 must be financed by **many lenders**.
 - Lenders are competitive and require expected return R .
- **Intermediary.**
 - The intermediary is risk neutral and has zero initial wealth.
 - It raises funds from depositors, lends to entrepreneurs, and monitors borrowers.
 - Depositors can observe only the intermediary's payments to them; they cannot directly observe monitored project outcomes or the intermediary's actions.

2.3 Project technology and returns

- Each project requires **one unit** of input today.
- End-of-period output is a random variable y bounded on

$$0 \leq y \leq \bar{y} < \infty.$$

- All agents agree on the distribution of y , and the project is socially worthwhile because

$$\mathbb{E}[y] > R + K.$$

- A crucial assumption is that the entrepreneur privately observes the realization of y .
- The realization $y = 0$ occurs with positive probability, so a fully riskless promise to repay R cannot be enforced without some penalty or monitoring device.

2.4 Monitoring technology

- By spending $K > 0$, a monitor can observe the true project outcome.
- Monitoring must be decided **before** the outcome is known.
- Monitoring information is **not automatically public**. If one lender or intermediary observes y , others do not automatically observe it as a by-product.
- This is what makes direct monitoring by many lenders wasteful and creates room for a monitoring specialist.

2.5 Contractual objects

- **Direct finance payment.**

- Let z denote the aggregate payment from the entrepreneur to lenders.
- Feasibility requires

$$0 \leq z \leq y.$$

- **Bankruptcy penalty.**

- If lenders cannot observe y , they can still threaten a **non-pecuniary penalty** $\phi(z)$ when repayment is low.
- Economically, this stands for bankruptcy costs, loss of reputation, time in reorganization, and related deadweight losses.

- **Intermediary loan repayments.**

- If the intermediary monitors entrepreneur i , it can design a payment schedule $g_i(y_i)$.
- Aggregate receipts of the intermediary are

$$G_N = \sum_{i=1}^N g_i(y_i).$$

- **Depositor repayment.**

- Let Z_N denote the intermediary's total payment to depositors.
- Feasibility requires $Z_N \leq G_N$.

2.6 Timing

- Date 0: entrepreneurs seek financing; lenders or the intermediary decide whether to monitor.
- Date 1: project outcomes are realized and privately observed by entrepreneurs.
- Date 2: monitored information is used to enforce contracts; entrepreneurs pay lenders or the intermediary; the intermediary pays depositors.

2.7 Key economic objects to keep track of

- $\mathbb{E}[\phi^*(y)]$: expected bankruptcy cost under direct finance without monitoring.
- mK : total cost if all m lenders directly monitor.
- $K + D_N$: total per-project cost under delegated monitoring, where D_N is the delegation cost.
- h : face value of the entrepreneur's direct debt contract.
- H_N : face value of the intermediary's debt to depositors.

3 Models and empirical specifications (all equations + interpretation)

3.1 Entrepreneur's direct contracting problem

The entrepreneur chooses a contract to maximize expected residual payoff subject to giving lenders competitive expected repayment:

$$\max_{\phi(\cdot)} \mathbb{E}_y \left[\max_{z \in [0, y]} \{y - z - \phi(z)\} \right]$$

subject to

$$z \in \arg \max_{z' \in [0, y]} \{y - z' - \phi(z')\}$$

and

$$\mathbb{E}_y[z] \geq R.$$

Interpretation.

- The entrepreneur privately observes y , so repayment must be incentive compatible.
- The only costlessly observable object is the payment z itself.
- Non-pecuniary penalties $\phi(z)$ are the device used to induce truthful repayment.

3.2 Proposition 1: the optimal direct contract is debt

The optimal penalty schedule is

$$\phi^*(z) = \max\{h - z, 0\},$$

where h is the smallest solution to

$$\mathbb{P}(y < h)\mathbb{E}[y \mid y < h] + \mathbb{P}(y \geq h)h = R.$$

Interpretation.

- This is exactly a **debt contract** with face value h .
- If output is high enough, the entrepreneur pays the fixed amount h .
- If output is too low, the entrepreneur pays all available cash and suffers a **bankruptcy penalty** equal to the shortfall $h - z$.
- So debt is optimal here because it is the **least costly incentive-compatible contract**. Any softer contract would not induce repayment; any harsher contract would waste more surplus.

3.3 Direct monitoring benchmark

If all m lenders monitor directly, the total monitoring cost is mK . Direct monitoring is worth doing only if

$$mK \leq \mathbb{E}[\phi^*(y)].$$

Interpretation.

- Monitoring eliminates bankruptcy penalties, but when lenders are numerous it duplicates effort.
- This is the basic inefficiency that creates a role for delegated monitoring.
- If no lender monitors because each one hopes others will do it, there is a free-rider problem; if all monitor, there is wasteful duplication.

3.4 Delegated monitoring by an intermediary

If the intermediary monitors N entrepreneurs, entrepreneur i pays $g_i(y_i)$, so aggregate intermediary receipts are

$$G_N = \sum_{i=1}^N g_i(y_i).$$

The intermediary then makes total payment Z_N to depositors and faces a non-pecuniary penalty if repayment is too low. The optimal penalty schedule is again debt-like:

$$\Phi_N(Z_N) = \max\{H_N - Z_N, 0\},$$

where H_N is the smallest solution to

$$\mathbb{P}(G_N < H_N)\mathbb{E}[G_N \mid G_N < H_N] + [1 - \mathbb{P}(G_N < H_N)]H_N = NR.$$

Interpretation.

- The intermediary has exactly the same kind of incentive problem as the entrepreneur: depositors cannot observe what it actually collected.
- Therefore the optimal intermediary–depositor contract is also **debt**.
- This is a major result of the paper: debt is optimal both at the borrower level and at the intermediary level.

3.5 Viability condition for intermediation

Let D denote the expected delegation cost per project. Delegated monitoring is socially worthwhile if

$$K + D \leq \min\{\mathbb{E}[\phi^*(y)], mK\}.$$

Interpretation.

- The intermediary must beat both alternatives:
 - direct debt finance with bankruptcy costs, and
 - direct monitoring by all lenders.
- The whole paper is about characterizing D and showing when diversification makes it small enough.

3.6 Why one monitored loan is not enough

If the intermediary monitors only one entrepreneur ($N = 1$), it is **not viable**.

Logic.

- With one loan, the intermediary’s cash inflow is no more secure than the entrepreneur’s own repayment stream.
- So the intermediary still needs essentially the same bankruptcy incentives as the entrepreneur.
- But on top of that, the intermediary also incurs the physical monitoring cost K .

Interpretation. A bank that makes only one monitored loan is not special. Intermediation becomes distinct only when the intermediary pools many loans.

3.7 Proposition 2: diversification drives delegation costs to zero

Diamond’s key construction is to choose loan contracts such that each monitored entrepreneur pays the intermediary an expected amount

$$\mathbb{E}[g_i(y_i)] = R + K + D_N,$$

where $D_N > 0$ is the per-project delegation cost.

Set the intermediary’s face value of debt to depositors equal to

$$H_N = N \left(R + \frac{D_N}{2} \right),$$

and let the intermediary pay

$$Z_N = \min\{G_N, H_N\}.$$

Then the intermediary's expected profit net of monitoring costs is

$$\mathbb{E}[G_N] - H_N - NK = N(R + K + D_N) - N\left(R + \frac{D_N}{2}\right) - NK = \frac{ND_N}{2} > 0.$$

Because $\mathbb{E}[G_N] > H_N$, the weak law of large numbers implies that

$$\mathbb{P}(G_N \leq H_N) \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Hence the probability that the intermediary actually hits the bankruptcy region goes to zero, so

$$D_N \rightarrow 0.$$

Interpretation.

- This is the core theorem of the paper.
- Diversification stabilizes the intermediary's aggregate cash flow.
- Once the intermediary is sufficiently diversified, a fixed promised repayment to depositors is almost always feasible, so the expected incentive cost of making the intermediary honest becomes negligible.

3.8 Corollary: common observable factors do not destroy the result

The paper extends the result to a setting where project returns load on observable aggregate factors:

$$y_i = \sum_{j=1}^M \beta_{ij} F_j + \varepsilon_i,$$

where the F_j are observable ex post and the idiosyncratic components ε_i are bounded and independent.

Interpretation.

- Observable common risks can either be hedged in futures markets or written into deposit contracts.
- The intermediary only needs to bear the non-observable residual risk for incentive reasons.
- Therefore the diversification result is fundamentally about eliminating **idiosyncratic** delegation risk, not about pretending aggregate shocks do not exist.

3.9 Capital structure implication

As N grows large, the intermediary's debt becomes nearly riskless and leverage becomes high. The paper summarizes this with the face value of intermediary debt

$$H(N) = N \left(R + \frac{D_N}{2} \right),$$

while the expected future value of the intermediary is

$$N(R + D_N + K).$$

Interpretation.

- The intermediary finances itself primarily with **debt-like claims**.
- Yet default is rare because diversification makes the promised payment very safe.
- This matches the basic structure of banks: lots of debt, very little equity, and low normal-state default probability.

3.10 Risk-aversion extension: when does diversification help?

To isolate the risk-sharing channel, Diamond studies a separate model without the non-negativity wealth constraint. For two independent bounded gambles x_1 and x_2 , define risk premia p_1, p_2, p_{1+2} by

$$\mathbb{E}[U(W_0 + x_i + p_i)] = U(W_0 + \mathbb{E}[x_i]), \quad i = 1, 2,$$

and

$$\mathbb{E}[U(W_0 + x_1 + x_2 + p_{1+2})] = U(W_0 + \mathbb{E}[x_1 + x_2]).$$

Adding risks lowers the per-risk premium when

$$p_{1+2} < p_1 + p_2.$$

Interpretation.

- This section distinguishes two kinds of diversification:
 - **subdividing risks**: many bankers share a portfolio of projects,
 - **adding risks**: one banker simply holds more independent projects.
- Subdividing risks always helps because it is genuine risk sharing.
- Adding risks helps only under special curvature conditions on utility; it does *not* help under constant absolute risk aversion.

3.11 Comparison with Leland–Pyle (1977)

The paper applies the same logic to a signalling model of entrepreneurial finance.

Main result.

- If one **single** banker with constant absolute risk aversion simply adds more independent projects, signalling costs do not fall.
- If instead **N bankers** share the risks of **N projects**, each banker's problem becomes equivalent to signalling a project whose specific-risk variance is only

$$\frac{\sigma^2}{N}.$$

- Therefore signalling costs fall with true internal diversification by risk subdivision.

Interpretation. The message is the same as in the main model: diversification helps when it changes the incentive problem **inside the intermediary organization**, not when outside investors merely hold a diversified set of intermediary claims.

4 Identification and instruments (what makes the estimates credible)

4.1 This is a theory paper: what pins down the results

- There is no empirical identification problem here. The relevant question is what **analytically pins down** the delegated-monitoring result.
- Four ingredients do the heavy lifting:
 1. **private information about project outcomes**,
 2. **costly monitoring** that is not publicly observable,
 3. **many dispersed lenders**, and
 4. **internal diversification** inside the intermediary.

4.2 Why private observability matters

- If project output y were publicly observable, simple state-contingent contracts could implement repayment without any bankruptcy penalties or delegated monitoring issues.
- The entire argument begins from the fact that the entrepreneur privately sees y , while outsiders see only transfers.

4.3 Why many lenders are essential

- With one lender, direct monitoring is straightforward.
- With many lenders, direct monitoring either duplicates effort or collapses under free riding.
- This is why the paper treats financial intermediation as a solution to a **many-principal agency problem**, not just a one-lender contracting problem.

4.4 Why the non-negativity constraint matters

- The entrepreneur cannot pay more than current output, so repayment promises are limited by realized cash flow.
- This is why debt with bankruptcy penalties appears: when output is low, the entrepreneur cannot simply make up the shortfall.
- The same logic reappears at the intermediary level when deposits are financed out of realized loan repayments.

4.5 Why debt emerges twice

- The entrepreneur-to-lender contract is debt because it minimizes expected deadweight penalties subject to incentive compatibility.
- The intermediary-to-depositor contract is also debt because depositors cannot observe what the intermediary collected.
- So debt is not assumed; it is derived as the cheapest feasible way to make transfers incentive compatible under limited observability.

4.6 Why diversification must occur inside the intermediary

- Depositors themselves could diversify across many intermediary claims, but that does *not* solve the intermediary's incentive problem.
- What matters is that the intermediary's own cash flow becomes less volatile, which reduces the probability of hitting the punishment region.
- This is why Diamond emphasizes that diversification **within** the intermediary cannot be replaced by diversification **across** intermediaries by principals.

4.7 Why bounded independent returns are enough for Proposition 2

- Bounded returns guarantee that extreme shortfalls cannot explode.
- Independence guarantees that average portfolio returns concentrate around their mean as the portfolio expands.
- Together, these conditions give the law-of-large-numbers result that drives delegation costs to zero.

4.8 Why common factors can be hedged away

- Risks that are **observable** do not need to be borne by the intermediary for incentive purposes.
- They can be hedged in futures markets or passed through to depositors using observable contingencies.
- The intermediary only needs to absorb the component of risk tied to private information and monitoring.

4.9 What the risk-aversion extension teaches

- The main model's diversification result does not rely on ordinary concave-utility diversification logic; it already works with universal risk neutrality.
- The risk-aversion extension clarifies a different point: **not all diversification is the same**.
- True internal risk sharing helps; merely piling more independent risks onto one balance sheet may not.

4.10 Limits of the model

- The paper is static and one-period.
- Monitoring is modeled in reduced form through a fixed cost K .
- Entrepreneurs are risk neutral in the main model.
- The intermediary is treated as a single agent for the baseline delegated-monitoring problem, with richer internal organization introduced only in later discussion.

These abstractions keep the mechanism extremely transparent: the paper is not trying to explain every feature of banking, but rather to isolate the incentive logic that makes intermediation possible.

5 Tables and figures

5.1 No empirical tables

- This paper is entirely theoretical. There are no regression tables, calibration tables, or empirical figures.
- The key objects to “read” are the propositions and comparative-static arguments.

5.2 Proposition 1: optimal direct finance uses debt

Read:

- Proposition 1 is the foundation of the whole paper.
- It shows that the optimal borrower–lender contract is not a complicated state-contingent mechanism; it is a simple debt contract with a bankruptcy penalty.
- This matters because the same debt logic is later applied to the intermediary itself.

Main takeaway. Debt is the cheapest way to induce repayment when the borrower privately observes output and cannot pay more than realized cash flow.

5.3 Section 2 direct-monitoring comparison

Read:

- The direct comparison is between:
 1. no monitoring with bankruptcy costs,
 2. all lenders monitoring directly at cost mK , and
 3. delegated monitoring through an intermediary at cost $K + D$.
- This is the cleanest way to understand where intermediation creates value.

Main takeaway. A bank is valuable only if it reduces total monitoring-related deadweight losses after accounting for its own incentive problem.

5.4 Proposition 2: diversification makes delegated monitoring viable

Read:

- Proposition 2 is the central theorem.
- It proves that the per-project delegation cost D_N goes to zero as the intermediary monitors more bounded independent loans.
- This is the paper's formal explanation for why diversified banks can exist even though depositors do not monitor them directly.

Main takeaway. Scale and diversification are not incidental features of banks; they are what make delegated monitoring incentive compatible.

5.5 Corollary to Proposition 2: observable aggregate risk does not break the theory

Read:

- The corollary allows project returns to share observable common factors.
- This makes the theory much more realistic: loan returns can co-move with interest rates, macro conditions, or input prices.
- Because those components are observable, they can be hedged or contractually passed through.

Main takeaway. The model does not require a literally idiosyncratic world. It requires that the non-observable residual risks be diversifiable.

5.6 Section 5 and the appendix: the Leland–Pyle comparison

Read:

- This part is conceptually important because it shows the paper's mechanism is not tied to one exact model of asymmetric information.
- It also distinguishes sharply between two notions of diversification:
 - **adding risks** to one risk-averse agent,
 - **subdividing risks** across many agents within the intermediary.

Main takeaway. Intermediation becomes viable when diversification reduces the incentive cost **inside the organization**; the paper is not claiming that any large portfolio automatically justifies an intermediary.

6 Short summary

- **Method.** A pure theory model of entrepreneurial borrowing, costly monitoring, and delegated monitoring through an intermediary.

- **Main contract result.** The optimal repayment contract under private output observability is **debt with bankruptcy penalties**.
- **Main intermediation result.** Delegated monitoring by a diversified intermediary is viable because the delegation cost per borrower, D_N , converges to zero as the number of independent loans grows.
- **Main lesson.** Financial intermediaries exist because diversification makes it cheap to give one agent the job of monitoring many borrowers on behalf of many depositors.

7 Conclusion

7.1 Core conclusion

- The paper shows that financial intermediation can be understood as an optimal organizational response to costly monitoring.
- Banks economize on duplicate monitoring by centralizing the monitoring task.
- Diversification makes that centralization incentive compatible, so the intermediary can honestly promise depositors a nearly fixed payment.

7.2 Economic intuition

- One monitored loan does not justify a bank because the bank then looks too much like the borrower it is monitoring.
- Many monitored loans do justify a bank because portfolio diversification makes the bank's aggregate cash flow predictable enough to back safe debt.
- The bank therefore “absorbs” private information on the asset side and issues simple claims on the liability side.

7.3 Broader implications

- **Capital structure.** Intermediaries should be highly leveraged and issue mostly safe debt-like claims.
- **Portfolio structure.** Intermediaries should prefer diversified portfolios of loans whose non-observable components are not too correlated.
- **Hedging.** Observable aggregate risks should be hedged or passed through, not borne unnecessarily by banks.
- **Asset illiquidity.** Monitored loans are naturally illiquid because selling them would require duplicating monitoring effort and may trigger adverse-selection problems.
- **Industrial organization of finance.** There are increasing returns to scale in intermediation because diversification reduces delegation costs, although those gains eventually flatten as D_N approaches zero.

7.4 Takeaway

Diamond's central insight is simple and durable: a bank is valuable because it can do something individual depositors cannot do cheaply on their own — **monitor borrowers once, pool the resulting cash flows, and issue simple debt claims against a diversified portfolio**. That is the microfoundation of financial intermediation in this paper.